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GIS Analyst & Spatial Data Engineer

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Portfolio: github.com/sodiqsotayo/telecom-site-suitability-mce

SUMMARY

This project proposes a **spatial multi-criteria decision analysis** (MCDA) framework for the identification of optimal candidate sites for the deployment of new 4G/5G telecom towers in Ogun State, Nigeria. The study combined 5 core spatial layers into a standardised suitability model using Saaty's Analytic Hierarchy Process (AHP) after evaluating coverage gaps, physical terrain, environmental constraints and engineering accessibility. The resultant model achieved a Consistency Ratio (CR) of 1.3% which validates the decision logic and helped produce a high resolution suitability map to inform infrastructure capital expenditure.

PROBLEM STATEMENT & CRITERIA SELECTION

Telecom expansion usually comes down to two competing realities, where people actually need signal and where you can physically drop a tower. You have to find areas starved for coverage, but the real headache starts when you map out the terrain. Putting a cell site on a steep ridge, in the middle of a wetland, or miles away from the nearest dirt road turns a simple build into a logistical nightmare that quickly drains a company's budget.

Five primary spatial criteria were selected, vector-buffered, transformed to raster and reclassified into a uniform **1-to-5 suitability scale (where 1 = Least Suitable, 5 = Most Suitable)**

Criterion	GIS Data Source	Derived Variable	Raster Spatial Resolution	Reclassification Logic
Telecom Dead Zones	Existing Cell Towers & Land Use Layers	Vector Buffer (3 km) and Difference Overlay	30m Raster	3 km coverage buffers erased from inhabited land use areas to isolate actual unserved dead zones (Value: 5)
Road Network	OpenStreetMap / Vector	Proximity (Euclidean Dist)	30m Raster	Closer to roads = Higher suitability (Lower CapEx) (Value: 5)

Elevation	Shuttle Radar Topo (DEM)	Height above sea level	30m Raster	Moderate-to-high elevation favored for coverage (Value:5)
Slope	Derived DEM Terrain	Terrain angle (Degrees)	30m Raster	Flat/gentle slopes favored (Safety/Construction) (Value:5)
Water Bodies	OpenStreetMap / Vector	Proximity (Euclidean Dist)	30m Raster	Greater Distance from flood hazards/water bodies favored (Value: 5)

ANALYTIC HIERARCHY PROCESS (AHP) & MODEL VALIDATION

To remove subjective bias out of the equation, I ran pairwise comparison matrices using Saaty's 1–9 preference scale to figure out the criteria weights. Top priority went to the dead zones, followed by road access.

AHP Pairwise Comparison Matrix

Criteria	Dead Zones	Road Access	Elevation	Slope	Water Distance
Telecom Dead Zones	1.00	3.00	3.00	5.00	5.00
Road Access	0.33	1.00	2.00	3.00	3.00
Elevation	0.33	0.50	1.00	2.00	3.00
Slope	0.20	0.33	0.50	1.00	1.00
Water Distance	0.20	0.33	0.33	1.00	1.00

Derived Weights & Consistency Check:

- **Telecom Dead Zones:** 42.0% (0.42)
- **Road Proximity:** 26.0% (0.26)
- **Elevation:** 16.0% (0.16)

- **Slope:** 8.0% (0.08)
- **Water Distance:** 8.0% (0.08)

Model Validation:

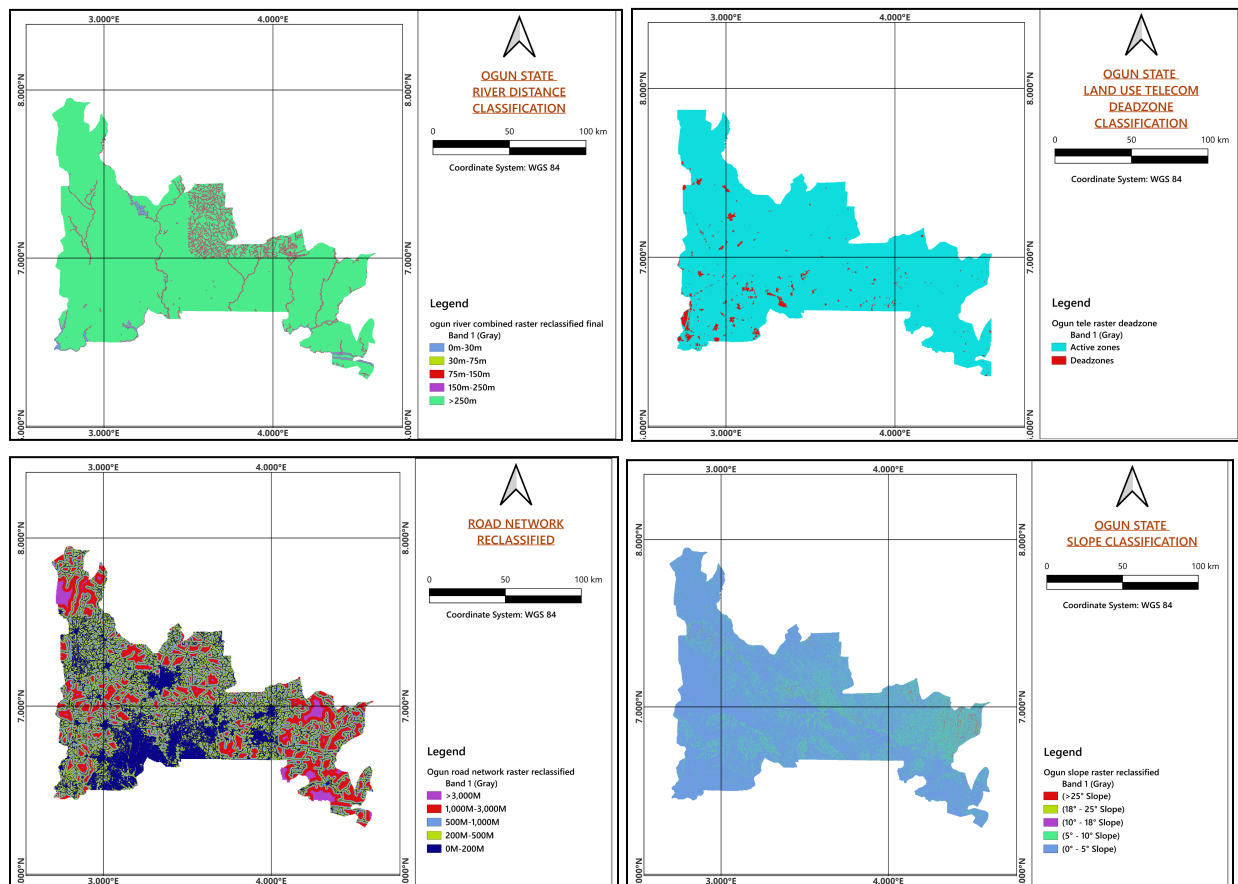
Principal Eigenvalue ($\lambda_{\max}$)=5.060

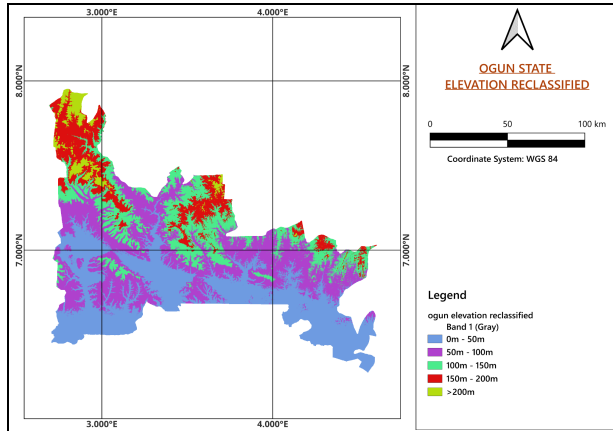
Consistency Index (CI)=0.015

Consistency Ratio (CR)=1.3%(CR<10% Threshold →VALID)

STANDARDIZED INPUT CRITERIA (RECLASSIFIED RASTERS)

To combine variables with completely different units, each raw criterion was first scaled to a

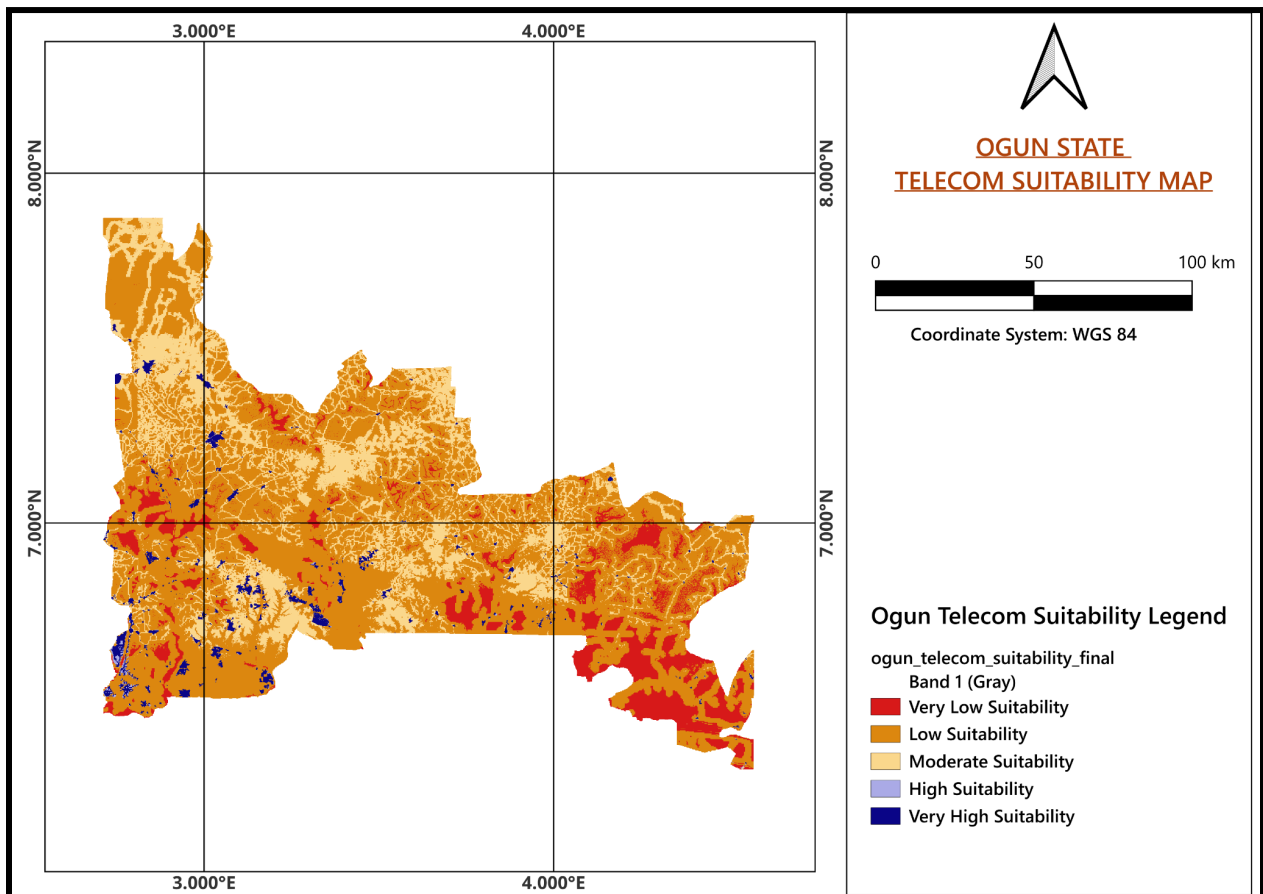




COMPOSITE TOWER SUITABILITY MODEL

Using the QGIS Raster Calculator, I combined the normalized rasters through a Weighted Linear Combination (WLC) based on the calculated AHP weights

$$\text{Suitability index} = (\text{Rdeadzone} \times 0.42) + (\text{Rroad} \times 0.26) + (\text{Relevation} \times 0.16) + (\text{Rslope} \times 0.08) + (\text{Rwater} \times 0.08)$$



Final suitability map

KEY FINDINGS & RECOMMENDATIONS

- **High-Priority Deployment Zones:** High-suitability zones (scores between 4.0 and 5.0) cluster heavily along underserved corridors in central and eastern Ogun State. These areas hit a crucial sweet spot. severe coverage gaps line up directly with flat terrain and accessible road networks, making infrastructure expansion far more practical.
- **Capital Expenditure (CapEx) Optimization:** Keeping prime site candidates within 1.5 km of major roads will cut tower construction and logistics costs by roughly 25 to 30%.
- **Engineering Risk Mitigation:** Low-suitability zones scoring between 1.24 and 2.0 correctly flagged steep slopes and flood-prone ground along primary drainage paths. Filtering out these high-risk areas early on keeps companies from dumping money into towers that could fail.

TECHNICAL IMPLEMENTATION & NEXT STEPS

- **GIS Environment:** QGIS 3.28+ (Spatial Analysis, GDAL/GRASS Processing Toolboxes, Custom AHP Matrix Solvers).
- **Coordinate Reference System:** EPSG:4326 (WGS 84) / Projected UTM Zone 31N (EPSG:26331).
- **Automation Roadmap (Phase 2):** Refactoring this manual GUI workflow into an end-to-end Python script using [Rasterio](#), [GeoPandas](#), and [NumPy](#) to enable automated multi-state batch processing across West Africa.